

```

(kali㉿kali)-[~]
$ mkdir -p ~/CIP-B102_Lab6_Data_Carving

(kali㉿kali)-[~]
$ cd ~/CIP-B102_Lab6_Data_Carving

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ wget -q https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd -O Ch01InChap01.dd

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_file/J_ub_law.jpg -O J_ub_law.jpg

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z -O 120M.7z

```

```

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ ls -lh
total 39M
-rw-rw-r-- 1 kali kali 36M Jul 12 06:41 120M.7z
-rw-rw-r-- 1 kali kali 1.5M Jul 12 06:23 Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.7M Jul 12 06:26 J_ub_law.jpg

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$

```

```

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ file Ch01InChap01.dd J_ub_law.jpg 120M.7z
Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID ", 'k0nIHC" cached by Windows 9M, root entries 224, sectors 2880 (volumes ≤ 32 MB), sectors/FAT 9, sectors/track 18, dos < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT
J_ub_law.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920×1080, components 3
120M.7z: 7-zip archive data, version 0.4

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ md5sum Ch01InChap01.dd J_ub_law.jpg 120M.7z
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ sha256sum Ch01InChap01.dd J_ub_law.jpg 120M.7z
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg
2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce 120M.7z

```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ xxd J_ub_law.jpg | head
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201  ....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  ....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201  ....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100  ....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00  ....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801  ....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100  ....1 ...!
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ xxd J_ub_law.jpg | tail
0019d160: ecfe 3d52 1918 3008 da88 b489 7f22 feae  ..=R..0....." ..
0019d170: 2def 629c 7ab5 16bc 7acd 5750 6ba9 63a7  -.b.z ... z.WPk.c.
0019d180: 9155 0524 f1a8 648f 4d6a f9c7 95e4 8981  .U.$..d.Mj.....
0019d190: 55d6 1519 8293 6360 41e0 fbd5 0530 3aa9  U.....c`A....0:.
0019d1a0: 0bab 554f 4d91 346a 25a7 99d9 2a52 a1bd  ..UOM.4j%...*R..
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80  )O.J.Y..(.QZ.v:..
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ xxd -l 120 J_ub_law.jpg
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ... ..Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ... .. .. ..@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201  .. .. .. .. ..
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  ... .. .. .. ..
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ... .. .. .. ..
00000050: 0000 1001 0200 0900 0000 e600 0000 1201  .. .. .. .. ..
00000060: 0300 0100 0000 0100 0000 1501 0300 0100  ... .. .. .. ..
00000070: 0000 0300 0000 1a01  .. .. .. ..
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ xxd -p J_ub_law.jpg > J_ub_law_hexdump.txt
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ grep -o "ffd8" J_ub_law_hexdump.txt | head
```

```
ffd8
ffd8
ffd8
ffd8
ffd8
ffd8
```



```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ xxd -r -p J_ub_law_hexdump.txt J_ub_law_reverse_dump.jpg

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ file J_ub_law_reverse_dump.jpg
J_ub_law_reverse_dump.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ md5sum J_ub_law.jpg J_ub_law_reverse_dump.jpg
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law_reverse_dump.jpg

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ sha256sum J_ub_law.jpg J_ub_law_reverse_dump.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law_reverse_dump.jpg
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ xxd -s 0x100 -l 0x9f J_ub_law.jpg
00000100: 2700 00c0 c62d 0010 2700 0041 646f 6265 ' .. ..-.. ' ..Adobe
00000110: 2050 686f 746f 7368 6f70 2032 312e 3120 Photoshop 21.1
00000120: 284d 6163 696e 746f 7368 2900 3230 3230 (Macintosh).2020
00000130: 3a30 383a 3230 2031 303a 3230 3a30 3500 :08:20 10:20:05.
00000140: 484f 5741 5244 204b 4f52 4e00 3200 3000 HOWARD KORN.2.0.
00000150: 3100 3300 2c00 2000 6300 6100 6d00 7000 1.3.,. .c.a.m.p.
00000160: 7500 7300 2c00 2000 4d00 6500 7200 7200 u.s.,. .M.e.r.r.
00000170: 6900 6300 6b00 2000 5300 6300 6800 6f00 i.c.k. .S.c.h.o.
00000180: 6f00 6c00 2000 6f00 6600 2000 4200 7500 o.l. .o.f. .B.u.
00000190: 7300 6900 6e00 6500 7300 7300 0000 00 s.i.n.e.s.s....
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ strings J_ub_law.jpg | head -50
Exif
NIKON CORPORATION
NIKON D4
Adobe Photoshop 21.1 (Macintosh)
2020:08:20 10:20:05
HOWARD KORN
0221
2013:10:08 18:56:21
2013:10:08 18:56:21
2027750
17.0-35.0 mm f/2.8
Adobe_CM
Adobe
b34r
7GWgw
AQaq"
dEU6te
'7GWgw
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ strings J_ub_law.jpg | grep -Ei "nikon|photoshop|adobe|date|copyright|author|name"
NIKON CORPORATION
NIKON D4
Adobe Photoshop 21.1 (Macintosh)
Adobe_CM
Adobe
"Photoshop 3.0
printerNameTEXT
Adobe_CM
Adobe
http://ns.adobe.com/xap/1.0/
" id="W5M0MpCehiHzreSzNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 6.0-c
002 79.164352, 2020/01/30-15:50:38 " > <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-s
yntax-ns#" > <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/aux/" xmlns:dc="h
ttp://purl.org/dc/elements/1.1/" xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="ht
tp://ns.adobe.com/xap/1.0/" xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns.ad
obe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#"
xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns:lr="http://ns.adobe.com/lightroom/1.
0/" aux:ApproximateFocusDistance="4294967295/1" aux:ImageNumber="115422" aux:Lens="17.0-35.0 mm f/2
.8" aux:LensID="99" aux:LensInfo="170/10 350/10 28/10 28/10" aux:SerialNumber="2027750" dc:format="
```

```
<?xpacket end="w"?>
Copyright 1999 Adobe Systems Incorporated
Adobe RGB (1998)
Adobe
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ █
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
```

Image Type: raw

Size in bytes: 1474560
Sector size: 512

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ mmls Ch01InChap01.dd
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ fsstat Ch01InChap01.dd | head -40
FILE SYSTEM INFORMATION
```

File System Type: FAT12

OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):

Sectors before file system: 0

File System Layout (in sectors)

Total Range: 0 - 2879

* Reserved: 0 - 0

** Boot Sector: 0

* FAT 0: 1 - 9

* FAT 1: 10 - 18

* Data Area: 19 - 2879

** Root Directory: 19 - 32

** Cluster Area: 33 - 2879

METADATA INFORMATION

Range: 2 - 45782

Root Directory: 2

CONTENT INFORMATION

Sector Size: 512

Cluster Size: 512

```
Cluster Size: 512
Total Cluster Range: 2 - 2848
```

```
FAT CONTENTS (in sectors)
```

```
33-236 (204) → EOF
285-311 (27) → EOF
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ fls Ch01InChap01.dd
```

```
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ fls -d Ch01InChap01.dd
```

```
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ icat Ch01InChap01.dd 12 > recovered_from_inode.bin
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ icat Ch01InChap01.dd 8 > Billing_Letter.doc

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ file Billing_Letter.doc
Billing_Letter.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10,
Code page: 1252, Title: 13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved
By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Ed
iting Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri Dec 9 14:
50:00 2005, Number of Pages: 1, Number of Words: 118, Number of Characters: 678, Security: 0

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ icat Ch01InChap01.dd 11 > confirmation.txt

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ file confirmation.txt
confirmation.txt: ASCII text, with CRLF line terminators
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ icat Ch01InChap01.dd 15 > letter1.txt

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ file letter1.txt
letter1.txt: ASCII text, with CRLF line terminators

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ cat confirmation.txt
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroi9

Thank you for your business

George
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ cat letter1.txt
```

Earl,

We need to meet on the 18th of August to confirm the work I am doing for you. Please contact me ASAP.

George

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ fls Ch01InChap01.dd
```

r/r 5: Client Info.mdb

r/r * 8: Billing Letter.doc

r/r * 11: confirmation.txt

r/r 13: Income.xls

r/r * 15: letter1.txt

r/r * 17: Regrets.doc

v/v 45779: \$MBR

v/v 45780: \$FAT1

v/v 45781: \$FAT2

V/V 45782: \$OrphanFiles

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]  
$ istat Ch01InChap01.dd 8
```

Directory Entry: 8

Not Allocated

File Attributes: File, Archive

Size: 24064

Name: _ILLIN~1.DOC

Directory Entry Times:

Written: 2005-12-09 06:50:28 (EST)

Accessed: 2005-12-09 00:00:00 (EST)

Created: 2005-12-09 06:59:05 (EST)

Sectors:

237 238 239 240 241 242 243 244
245 246 247 248 249 250 251 252
253 254 255 256 257 258 259 260
261 262 263 264 265 266 267 268
269 270 271 272 273 274 275 276
277 278 279 280 281 282 283

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]  
$ istat Ch01InChap01.dd 11
```

Directory Entry: 11

Not Allocated

File Attributes: File, Archive

Size: 227

Name: _ONFIR~1.TXT

Directory Entry Times:

Written: 2005-12-09 06:52:58 (EST)

Accessed: 2005-12-09 00:00:00 (EST)

Created: 2005-12-09 06:59:06 (EST)

Sectors:

284

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ istat Ch01InChap01.dd 15
```

```
Directory Entry: 15
```

```
Not Allocated
```

```
File Attributes: File, Archive
```

```
Size: 121
```

```
Name: _ETTER1.TXT
```

```
Directory Entry Times:
```

```
Written:      2005-12-09 06:51:50 (EST)
```

```
Accessed:     2005-12-09 00:00:00 (EST)
```

```
Created:      2005-12-09 06:59:09 (EST)
```

```
Sectors:
```

```
312
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ istat Ch01InChap01.dd 17
```

```
Directory Entry: 17
```

```
Not Allocated
```

```
File Attributes: File, Archive
```

```
Size: 23552
```

```
Name: _EGRETS.DOC
```

```
Directory Entry Times:
```

```
Written:      2005-12-09 06:50:52 (EST)
```

```
Accessed:     2005-12-09 00:00:00 (EST)
```

```
Created:      2005-12-09 06:59:09 (EST)
```

```
Sectors:
```

```
313 314 315 316 317 318 319 320
```

```
321 322 323 324 325 326 327 328
```

```
329 330 331 332 333 334 335 336
```

```
337 338 339 340 341 342 343 344
```

```
345 346 347 348 349 350 351 352
```

```
353 354 355 356 357 358
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ ls -lh File_carving.docx
ls: cannot access 'File_carving.docx': No such file or directory

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ binwalk J_ub_law.jpg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, EXIF standard
12	0xC	TIFF image data, little-endian offset of first image directory: 8
26844	0x68DC	Copyright string: "Copyright 1999 Adobe Systems Incorporated"

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ file J_ub_law.jpg
J_ub_law.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ binwalk Ch01InChap01.dd
```

DECIMAL	HEXADECIMAL	DESCRIPTION
---------	-------------	-------------

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ binwalk -e Ch01InChap01.dd
```

DECIMAL	HEXADECIMAL	DESCRIPTION
---------	-------------	-------------

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ ls -lah _Ch01InChap01.dd.extracted
ls: cannot access '_Ch01InChap01.dd.extracted': No such file or directory
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ file Ch01InChap01.dd
Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID "", 'k0nIHC' cached by Windows 9M, root entries 224, sectors 2880 (volumes ≤ 32 MB), sectors/FAT 9, sectors/track 18, dos < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ img_stat Ch01InChap01.dd
```

IMAGE FILE INFORMATION
Image Type: raw
Size in bytes: 1474560
Sector size: 512

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ mmls Ch01InChap01.dd
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ fls -r Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ binwalk 120M.7z
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	7-zip archive data, version 0.4
1396360	0x154E88	Zip archive data, at least v1.0 to extract, compressed size: 190030, uncompressed size: 190030, name: ppt/fonts/font4.fntdata
1586455	0x183517	Zip archive data, at least v1.0 to extract, compressed size: 18171, uncompressed size: 18171, name: ppt/fonts/font5.fntdata
1604679	0x187C47	Zip archive data, at least v1.0 to extract, compressed size: 34163, uncompressed size: 34163, name: ppt/fonts/font6.fntdata
1638898	0x1901F2	Zip archive data, at least v1.0 to extract, compressed size: 172668, uncompressed size: 172668, name: ppt/fonts/font7.fntdata
2445986	0x2552A2	Zip archive data, at least v1.0 to extract, compressed size: 88758, uncompressed size: 88758, name: ppt/media/image14.png
3666842	0x37F39A	JB00T STAG header, image id: 0, timestamp 0x1EFB070E, image size: 2655696701 bytes, image JB00T checksum: 0xFFDC, header JB00T checksum: 0x9300
23522902	0x166EE56	JB00T SCH2 kernel header, compression type: none, Entry Point: 0xEE8895A, image size: 1608501141 bytes, data CRC: 0x6977FA7D, Data Address: 0x24241959, rootfs offset: 0x88BA0829, rootfs size: 1182421830 bytes, rootfs CRC: 0xD70FC9DD, header CRC: 0x3ECF7B1F, header s

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ ls -lh 120M.7z
```

```
-rw-rw-r-- 1 kali kali 36M Jul 12 06:41 120M.7z
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ binwalk 120M.7z
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	7-zip archive data, version 0.4
1396360	0x154E88	Zip archive data, at least v1.0 to extract, compressed size: 190030, uncompressed size: 190030, name: ppt/fonts/font4.fntdata
1586455	0x183517	Zip archive data, at least v1.0 to extract, compressed size: 18171, uncompressed size: 18171, name: ppt/fonts/font5.fntdata
1604679	0x187C47	Zip archive data, at least v1.0 to extract, compressed size: 34163, uncompressed size: 34163, name: ppt/fonts/font6.fntdata
1638898	0x1901F2	Zip archive data, at least v1.0 to extract, compressed size: 172668, uncompressed size: 172668, name: ppt/fonts/font7.fntdata
2445986	0x2552A2	Zip archive data, at least v1.0 to extract, compressed size: 88758, uncompressed size: 88758, name: ppt/media/image14.png
3666842	0x37F39A	JB00T STAG header, image id: 0, timestamp 0x1EFB070E, image size: 265

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ file 120M.7z
```

```
120M.7z: 7-zip archive data, version 0.4
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ binwalk -e 120M.7z
```

DECIMAL	HEXADECIMAL	DESCRIPTION
---------	-------------	-------------

WARNING: Extractor.execute failed to run external extractor 'jar xvf '%e%': [Errno 2] No such file or directory: 'jar', 'jar xvf '%e%' might not be installed correctly

1396360	0x154E88	Zip archive data, at least v1.0 to extract, compressed size: 190030, uncompressed size: 190030, name: ppt/fonts/font4.fntdata
1586455	0x183517	Zip archive data, at least v1.0 to extract, compressed size: 18171, uncompressed size: 18171, name: ppt/fonts/font5.fntdata
1604679	0x187C47	Zip archive data, at least v1.0 to extract, compressed size: 34163, uncompressed size: 34163, name: ppt/fonts/font6.fntdata
1638898	0x1901F2	Zip archive data, at least v1.0 to extract, compressed size: 172668, uncompressed size: 172668, name: ppt/fonts/font7.fntdata
2445986	0x2552A2	Zip archive data, at least v1.0 to extract, compressed size: 88758, u

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ ls -lah _120M.7z.extracted
```

```
total 44M
```

drwxrwxr-x	2	kali	kali	4.0K	Jul 12 07:33	.
drwxrwxr-x	3	kali	kali	4.0K	Jul 12 07:33	..
-rw-rw-r--	1	kali	kali	34M	Jul 12 07:33	154E88.zip
-rw-rw-r--	1	kali	kali	12K	Jul 12 07:33	2085526
-rw-rw-r--	1	kali	kali	2.5M	Jul 12 07:33	2085526.zlib
-rw-rw-r--	1	kali	kali	11K	Jul 12 07:33	208825A
-rw-rw-r--	1	kali	kali	2.5M	Jul 12 07:33	208825A.zlib
-rw-rw-r--	1	kali	kali	11K	Jul 12 07:33	208A93F
-rw-rw-r--	1	kali	kali	2.5M	Jul 12 07:33	208A93F.zlib
-rw-rw-r--	1	kali	kali	17K	Jul 12 07:33	208B5C8
-rw-rw-r--	1	kali	kali	2.5M	Jul 12 07:33	208B5C8.zlib

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ 7z l 120M.7z

7-Zip 26.01 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-04-27
64-bit locale=en_US.UTF-8 Threads:32 OPEN_MAX:4096, ASM
```

```
Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)
```

```
Listing archive: 120M.7z
```

```
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1
```

Date	Time	Attr	Size	Compressed	Name
2021-09-22	22:59:31	D...A	0	0	120M
2021-09-22	22:59:31	A	124780544	36720256	120M/usb_fat_carving.001
2021-09-22	22:59:32	A	1685		120M/usb_fat_carving.001.txt
2021-09-22	22:59:32		124782229	36720256	2 files, 1 folders

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ 7z e 120M.7z

7-Zip 26.01 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-04-27
64-bit locale=en_US.UTF-8 Threads:32 OPEN_MAX:4096, ASM
```

```
Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)
```

```
Extracting archive: 120M.7z
```

```
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
```

```
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1
```

Everything is Ok

```
Folders: 1
Files: 2
Size:      124782229
Compressed: 36720470
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ ls -lh
```

total 166M

```
drwxrwxr-x 2 kali kali 4.0K Sep 22 2021 120M
-rw-rw-r-- 1 kali kali 36M Jul 12 06:41 120M.7z
drwxrwxr-x 2 kali kali 4.0K Jul 12 07:33 _120M.7z.extracted
-rw-rw-r-- 1 kali kali 24K Jul 12 07:07 Billing_Letter.doc
-rw-rw-r-- 1 kali kali 1.5M Jul 12 06:23 Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 227 Jul 12 07:07 confirmation.txt
```

```
-rw-rw-r-- 1 kali kali 3.3M Jul 12 06:47 J_ub_law_hexdump_singleline.txt
-rw-rw-r-- 1 kali kali 3.3M Jul 12 06:46 J_ub_law_hexdump.txt
-rw-rw-r-- 1 kali kali 1.7M Jul 12 06:26 J_ub_law.jpg
-rw-rw-r-- 1 kali kali 1.7M Jul 12 06:48 J_ub_law_reverse_dump.jpg
-rw-rw-r-- 1 kali kali 121 Jul 12 07:08 letter1.txt
-rw-rw-r-- 1 kali kali 0 Jul 12 07:03 recovered_from_inode.bin
-rw-rw-r-- 1 kali kali 119M Sep 22 2021 usb_fat_carving.001
-rw-rw-r-- 1 kali kali 1.7K Sep 22 2021 usb_fat_carving.001.txt
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ file * | tee file_types_lab6.txt
```

120M: directory

120M.7z: 7-zip archive data, version 0.4

_120M.7z.extracted: directory

Billing_Letter.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title: 13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 118, Number of Characters: 678, Security: 0

Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID ", 'k0nIHC" cached

```
by Windows 9M, root entries 224, sectors 2880 (volumes ≤32 MB), sectors/FAT 9, sectors/track 18, d
os < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT
confirmation.txt:          ASCII text, with CRLF line terminators
J_ub_low_hexdump_singleline.txt: ASCII text, with very long lines (65536), with no line terminators
J_ub_low_hexdump.txt:       ASCII text
J_ub_low.jpg:               JPEG image data, Exif standard: [TIFF image data, little-endian, d
irentries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON C
ORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920×1080, components 3
J_ub_low_reverse_dump.jpg:  JPEG image data, Exif standard: [TIFF image data, little-endian, d
irentries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON C
ORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920×1080, components 3
letter1.txt:               ASCII text, with CRLF line terminators
recovered_from_inode.bin:   empty
usb_fat_carving.001:        DOS/MBR boot sector MS-MBR Windows 7 english at offset 0×163 "Inva
lid partition table" at offset 0×17b "Error loading operating system" at offset 0×19a "Missing oper
ating system", disk signature 0×a1159a00; partition 1 : ID=0×e, active, start-CHS (0×0,2,3), end-CH
S (0×e,254,63), startsector 128, 243584 sectors
usb_fat_carving.001.txt:    Unicode text, UTF-8 (with BOM) text, with CRLF line terminators
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ md5sum * | tee md5_after_extract_lab6.txt
md5sum: 120M: Is a directory
md5sum: _120M.7z.extracted: Is a directory
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z
9fe241d0dde27e83442010b3eee5ad32 Billing_Letter.doc
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd
18e391549e4a8bc990b264f590fb33bb confirmation.txt
0653eae91c3bc8217f8b1d4bd39007e7 file_types_lab6.txt
7b99a4535ee37a9cfaa19a36b59b63a4 J_ub_low_hexdump_singleline.txt
a790fbdee8bcc5d30dd9c60119fcd7d6 J_ub_low_hexdump.txt
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_low.jpg
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_low_reverse_dump.jpg
49f68104660a091a4c015e86f3151237 letter1.txt
d41d8cd98f00b204e9800998ecf8427e recovered_from_inode.bin
ba4a1d0ba49f4a6667b00a3b3e85e604 usb_fat_carving.001
568d6b183c03ff6f3e01e3915846be60 usb_fat_carving.001.txt
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ IMAGE="usb_fat_carving.001"

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ ls -lh "$IMAGE"
-rw-rw-r-- 1 kali kali 119M Sep 22  2021 usb_fat_carving.001

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ img_stat "$IMAGE"
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 124780544
Sector size:    512
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ mmls "$IMAGE"
```

DOS Partition Table

Offset Sector: 0

Units are in 512-byte sectors

	Slot	Start	End	Length	Description
000:	Meta	00000000000	00000000000	00000000001	Primary Table (#0)
001:	_____	00000000000	00000000127	00000000128	Unallocated
002:	000:000	00000000128	0000243711	0000243584	Win95 FAT16 (0x0e)

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ fls -o 128 "$IMAGE"
```

r/r 3: USB (Volume Label Entry)

d/d 6: System Volume Information

r/r * 7: _est

r/r 10: .dropbox.device

d/d * 13: old_File_Carving_files

r/r * 15: B_ub_poe4.bmp

r/r * 18: B_zoom-eubie-mono.bmp

r/r * 21: Ballardlab8.java

r/r * 24: brittLab10.java

r/r * 24: brittLab10.java

r/r * 27: DO_example.doc

r/r * 30: DO_example2.doc

r/r * 33: G_BuiltForThis.gif

r/r * 35: G_zoom-sc.gif

r/r * 37: H_Form.html

r/r * 39: H_hello.html

r/r * 41: J_ub_law.jpg

r/r * 44: J_ub_night.jpg

r/r * 47: nps-2008-jean_outlook.pst

r/r * 50: P_CAS-zoom-6.png

r/r * 53: P_MSB_1_zoom.png

r/r * 57: pd_Evidence_search_techniques.pdf

r/r * 61: pd_Forensic_Report_Template.pdf

r/r * 64: pp_Number_Systems.pptx

r/r * 67: pp_one_page.pptx

r/r 69: readme.docx

r/r 70: readme.txt

r/r * 71: _tf_1.rtf

r/r * 74: T_Eubie-iphone-5-8.tiff

r/r * 78: T_youknowus-iphone-5-8.tiff

r/r * 81: W_axloadcomplete.wav

```
r/r * 81:      W_axloadcomplete.wav
r/r * 84:      W_speaker_test_sound.wav
r/r * 86:      Z_file.zip
r/r * 88:      江莎莎行踪轨迹.doc
r/r * 91:      nps-2008-jean_outlook.pst
v/v 3889667:   $MBR
v/v 3889668:   $FAT1
v/v 3889669:   $FAT2
V/V 3889670:   $OrphanFiles
```

```
(kaliⓈkali)-[~/CIP-B102_Lab6_Data_Carving]
```

```
$ fls -o 128 -d "$IMAGE"
```

```
r/r * 7:       _est
d/d * 13:      old_File_Carving_files
r/r * 15:      B_ub_poe4.bmp
r/r * 18:      B_zoom-eubie-mono.bmp
r/r * 21:      Ballardlab8.java
r/r * 24:      brittLab10.java
r/r * 27:      DO_example.doc
r/r * 30:      DO_example2.doc
r/r * 33:      G_BuiltForThis.gif
r/r * 35:      G_zoom-sc.gif
```

```
r/r * 35:      G_zoom-sc.gif
r/r * 37:      H_Form.html
r/r * 39:      H_hello.html
r/r * 41:      J_ub_law.jpg
r/r * 44:      J_ub_night.jpg
r/r * 47:      nps-2008-jean_outlook.pst
r/r * 50:      P_CAS-zoom-6.png
r/r * 53:      P_MSB_1_zoom.png
r/r * 57:      pd_Evidence_search_techniques.pdf
r/r * 61:      pd_Forensic_Report_Template.pdf
r/r * 64:      pp_Number_Systems.pptx
r/r * 67:      pp_one_page.pptx
r/r * 71:      _tf_1.rtf
r/r * 74:      T_Eubie-iphone-5-8.tiff
r/r * 78:      T_youknowus-iphone-5-8.tiff
r/r * 81:      W_axloadcomplete.wav
r/r * 84:      W_speaker_test_sound.wav
r/r * 86:      Z_file.zip
r/r * 88:      江莎莎行踪轨迹.doc
r/r * 91:      nps-2008-jean_outlook.pst
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ sudo cp /etc/scalpel/scalpel.conf /etc/scalpel/scalpel.conf.bak
[sudo] password for kali:

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ ls -lh /etc/scalpel/scalpel.conf*
-rw-r--r-- 1 root root 8.5K Sep 27  2025 /etc/scalpel/scalpel.conf
-rw-r--r-- 1 root root 8.5K Jul 12 08:25 /etc/scalpel/scalpel.conf.bak

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ sudo nano /etc/scalpel/scalpel.conf
```

```
#
#
# AOL ART files
#   art      y      150000  \x4a\x47\x04\x0e      \xcf\xcb\xcb
#   art      y      150000  \x4a\x47\x03\x0e      \xd0\xcb\x00\x00
#
# GIF and JPG files (very common)
#   gif      y      5000000  \x47\x49\x46\x38\x37\x61      \x00\x3b
#   gif      y      5000000  \x47\x49\x46\x38\x39\x61      \x00\x3b
#   jpg      y      5242880  \xff\xd8\xff???Exif      \xff\xd9
#   jpg      y      5242880  \xff\xd8\xff???JFIF      \xff\xd9
#
#
# PNG
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ sudo nano /etc/scalpel/scalpel.conf

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ IMAGE="usb_fat_carving.001"

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ ls -lh "$IMAGE"
-rw-rw-r-- 1 kali kali 119M Sep 22  2021 usb_fat_carving.001

(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ rm -rf output_lab6_scalpel
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ scalpel "$IMAGE" -o output_lab6_scalpel
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/kali/CIP-B102_Lab6_Data_Carving/usb_fat_carving.001"

Image file pass 1/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" →
8 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" →
9 files
Carving files from image.
Image file pass 2/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
```

```
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 0 seconds.
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ find output_lab6_scalpel -type f -maxdepth 3 -print
find: warning: you have specified the global option -maxdepth after the argument -ty
pe, but global options are not positional, i.e., -maxdepth affects tests specified b
efore it as well as those specified after it. Please specify global options before
other arguments.
output_lab6_scalpel/audit.txt
output_lab6_scalpel/jpg-0-0/00000000.jpg
output_lab6_scalpel/jpg-0-0/00000002.jpg
output_lab6_scalpel/jpg-0-0/00000001.jpg
output_lab6_scalpel/jpg-0-0/00000006.jpg
output_lab6_scalpel/jpg-0-0/00000007.jpg
output_lab6_scalpel/jpg-0-0/00000004.jpg
output_lab6_scalpel/jpg-0-0/00000005.jpg
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ cat output_lab6_scalpel/audit.txt | head -80

Scalpel version 1.60 audit file
Started at Sun Jul 12 08:31:40 2026
Command line:
scalpel usb_fat_carving.001 -o output_lab6_scalpel

Output directory: /home/kali/CIP-B102_Lab6_Data_Carving/output_lab6_scalpel
Configuration file: /etc/scalpel/scalpel.conf

Opening target "/home/kali/CIP-B102_Lab6_Data_Carving/usb_fat_carving.001"

The following files were carved:
File                Start                Chop                Length                Extracted Fr
om
00000010.jpg        3796992                NO                3148500                usb_fat_carv
ing.001
00000009.jpg        425822                 NO                4875802                usb_fat_carv

00000007.jpg        21358592               NO                5183267                usb_fat_carv
ing.001
00000016.jpg        36671488               NO                2889262                usb_fat_carv
ing.001
00000015.jpg        36571136               NO                2989614                usb_fat_carv
ing.001
00000014.jpg        36393610               NO                3167140                usb_fat_carv
ing.001
00000013.jpg        36352448               NO                3208302                usb_fat_carv
ing.001
00000012.jpg        35590996               NO                3969754                usb_fat_carv
ing.001
00000011.jpg        35350242               NO                4210508                usb_fat_carv
ing.001

Completed at Sun Jul 12 08:31:40 2026
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ find output_lab6_scalpel -type f -exec file {} \;
output_lab6_scalpel/audit.txt: ASCII text
output_lab6_scalpel/jpg-0-0/00000000.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:03 14:35:44], baseline, precision 8, 1920x1080, components 3
output_lab6_scalpel/jpg-0-0/00000002.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=12, width=16550, height=9313, bps=0, PhotometricInterpretation=RGB, orientation=upper-left]
output_lab6_scalpel/jpg-0-0/00000001.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:03 11:54:50], baseline, precision 8, 1920x1080, components 3
output_lab6_scalpel/jpg-0-0/00000006.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:23 15:24:33], baseline, precision 8, 1920x1080, components 3
output_lab6_scalpel/jpg-0-0/00000007.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:23 15:24:33], baseline, precision 8, 1920x1080, components 3
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ find output_lab6_scalpel -type f -exec md5sum {} \; > carved_files_md5.txt
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ cat carved_files_md5.txt
e7164d558441306a3f2bd72f78776c31 output_lab6_scalpel/audit.txt
280d788eb3d1d48b93e88374440b1cc9 output_lab6_scalpel/jpg-0-0/00000000.jpg
4a5d89f1d59cc27959925a6a93fb8ed8 output_lab6_scalpel/jpg-0-0/00000002.jpg
e553e2ec5b6d2c67829a0edbcc7609cd output_lab6_scalpel/jpg-0-0/00000001.jpg
23aebb41d1f892897e275e13145fead0 output_lab6_scalpel/jpg-0-0/00000006.jpg
80d6a778d127d09b03f01dc0821b0907 output_lab6_scalpel/jpg-0-0/00000007.jpg
85f5990b4cc8910d16c9a80b04f919a4 output_lab6_scalpel/jpg-0-0/00000004.jpg
8d6c3778abdb79fd36e07a79bd6abcc1 output_lab6_scalpel/jpg-0-0/00000005.jpg
d143fe524b792d62b55ad6b7e12c1dfa output_lab6_scalpel/jpg-0-0/00000003.jpg
aec4abf7cb00ca5fcf54967d454a801d output_lab6_scalpel/jpg-1-0/00000012.jpg
41cb821d0e0657ed0d2fabe09d685dfc output_lab6_scalpel/jpg-1-0/00000010.jpg
6ea7a6c2e3988a8147e58caee15dc9d6 output_lab6_scalpel/jpg-1-0/00000008.jpg
```

```
(kali㉿kali)-[~/CIP-B102_Lab6_Data_Carving]
$ find output_lab6_scalpel -type f \( -name "*.jpg" -o -name "*.JPG" \)
output_lab6_scalpel/jpg-0-0/00000000.jpg
output_lab6_scalpel/jpg-0-0/00000002.jpg
output_lab6_scalpel/jpg-0-0/00000001.jpg
output_lab6_scalpel/jpg-0-0/00000006.jpg
output_lab6_scalpel/jpg-0-0/00000007.jpg
output_lab6_scalpel/jpg-0-0/00000004.jpg
output_lab6_scalpel/jpg-0-0/00000005.jpg
output_lab6_scalpel/jpg-0-0/00000003.jpg
output_lab6_scalpel/jpg-1-0/00000012.jpg
output_lab6_scalpel/jpg-1-0/00000010.jpg
output_lab6_scalpel/jpg-1-0/00000008.jpg
output_lab6_scalpel/jpg-1-0/00000014.jpg
output_lab6_scalpel/jpg-1-0/00000016.jpg
output_lab6_scalpel/jpg-1-0/00000009.jpg
output_lab6_scalpel/jpg-1-0/00000015.jpg
```